ACME METALS

Digital Material Passport

ID **DMP-METAL-004** - *Version* **1.0.0**
Issue Date **2025-05-16** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	**Advanced Nuclear Systems Ltd.** Energy Square 456 Abingdon OX14 3DB GB procurement@advnuclear.example.com	*Order ID* **PO-56789** - Pos. 2 - 2025-04-10 - Quantity **200 kg** *Delivery ID* **DN-12345** - Pos. 1 - 2025-05-15 - Quantity **200 kg**

Product

Product Name	Stainless Steel 316L	*Number (UNS)*	S31603
Batch ID	H-87654-01	*Name (EN)*	1.4404
Surface Condition	2B	*Shape*	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm
Weight	200 kg	*Marking*	316L stamped on corner with medium depth and good legibility.
Production Date	2025-05-14	*Coloring*	Natural finish with full coverage for protection.
Country of Origin	DE	*Product Norm*	ASTM A240 (2023)

Chemical Analysis

Heat Number **H-87654** - *Melting Process* **EAF+AOD+LF** - *Casting Method* **ContinuousCasting** - *Casting Date* **2025-05-13** - *Sample Location* **Ladle**

Symbol	*Unit*	*Min*	*Max*	*Actual*
C	%		0.03	0.018
Cr	%	16	18	17.2
Ni	%	10	14	10.5
Mo	%	2	3	2.15
Mn	%		2	1.4

Symbol	*Unit*	*Min*	*Max*	*Actual*
Si	%		0.75	0.38
P	%		0.045	0.025
S	%		0.03	0.002
N	%		0.1	0.052

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Tensile Strength[1]	Rm	578 / 580 / 582 MPa - Average 580 - Min 578 - Max 582	515 MPa		ASTM E8	✓
0.2% Yield Strength[1]	Rp0.2	238 / 240 / 242 MPa - Average 240 - Min 238 - Max 242 - Std Dev 2	205 MPa		ASTM E8	✓
Elongation[1]	A	51 / 52 / 53 % - Average 52 - Min 51 - Max 53	40 %		ASTM E8	✓

[1] 3 specimens tested

Supplementary Tests

Test	*Result / limits*	*Method*	*Status*
Intergranular Corrosion - Resistance	Yes - No evidence of intergranular attack	ASTM A262 Practice E	✓
Pitting Corrosion Resistance 72 hours at 22°C in 6% $FeCl_3$	1.2 g/m² - max 4 g/m²	ASTM G48 Method A	✓
Crevice Corrosion Resistance 72 hours in 3.5% NaCl solution	Yes - No visible crevice corrosion	ASTM G78	✓
Stress Corrosion Cracking - Resistance Boiling 42% $MgCl_2$ solution, 100 hours	Yes - No cracking observed	ASTM G36	✓
Ferrite Content	2.5 % - max 5 %	ASTM A800	✓
Grain Size	7 ASTM No. - min 5 ASTM No.	ASTM E112	✓
Inclusion Rating Worst field rating	A1, B1, C1, D1 - max A2, B2, C2, D2	ASTM E45 Method A	✓
Ultrasonic Examination	Yes - No recordable indications	ASTM A388	✓
Liquid Penetrant Examination	Yes - No relevant indications	ASTM E165	✓
Weldability Test	0.4 mm - max 1 mm	Varestraint Test	✓
Surface Finish	25 µin Ra - max 32 µin Ra	ASME BPE SF1	✓

PREN (Pitting Resistance - Equivalent Number) Calculated using formula: %Cr + 3.3 × %Mo + 16 × %N	25.8 - min 24		✓
Dimensional Tolerance	-0.3 - 0.2 mm - target -0.4 - 0.4 mm	ASTM A480	✓
Flatness	4 mm/m - max 9 mm/m	ASTM A480	✓
PMI (Positive Material - Identification)	Yes - Material confirmed as 316L stainless steel	XRF Analysis	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with ASTM A240/A240M and meets all specified requirements. This material is suitable for nuclear applications in accordance with RCC-M code.

Validated by **Thomas Wagner**, Metallurgist, Quality Assurance - 2025-05-16

Validated by **Anna Schmidt**, Quality Manager, Quality Assurance - 2025-05-16